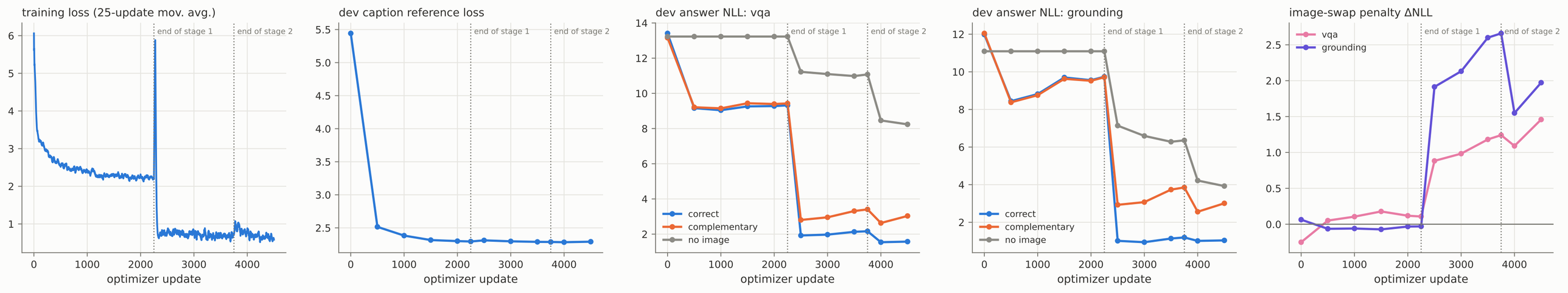
training loss (25-update mov. avg.)
end of stage 1
end of stage 2
optimizer update
dev caption reference loss
end of stage 1
end of stage 2
optimizer update
dev answer NLL: vqa
end of stage 1
end of stage 2
correct
complementary
no image
optimizer update
dev answer NLL: grounding
end of stage 1
end of stage 2
correct
complementary
no image
optimizer update
image-swap penalty ΔNLL
vqa
grounding
end of stage 1
end of stage 2
optimizer update